

Implementing Neural Networks in TensorFlow

TensorFlow is one of the most popular open-source frameworks used to build and train deep learning models. This course will focus on TensorFlow for its implementations (an alternative framework is PyTorch).

We will use TensorFlow to implement **inference** (the process of making a prediction) by defining the layers of our network.

The Dense Layer

In TensorFlow, the standard, fully-connected layer we have been studying is called a **Dense** layer.

- **Definition:** A Dense layer means that every neuron in that layer is connected to every activation from the previous layer.
- **Syntax:** You define a layer using `Dense(units=..., activation=...)`
 - **units:** The number of neurons in the layer.
 - **activation:** The activation function to be used by the neurons in that layer (e.g., 'sigmoid').

Forward Propagation in TensorFlow

The process of feeding an input x through the layers to compute the final output is called **forward propagation**.

Example 1: Coffee Roasting (2-layer network)

- **Goal:** Predict if coffee is good (1) or bad (0) based on 2 features.
- **Input (Layer 0):** $\vec{x} = [200, 17]$ (temperature, duration)

```
1 # Layer 1 (Hidden Layer)
2 # Define the layer's structure (3 neurons, sigmoid)
3 Layer1 = Dense(units=3, activation='sigmoid')
4 # Compute the layer's output (forward prop)
5 a1 = Layer1(x) # a1 will be a vector with 3 activations
6
7 # Layer 2 (Output Layer)
8 # Define the layer's structure (1 neuron, sigmoid)
9 Layer2 = Dense(units=1, activation='sigmoid')
10 # Compute the final output
11 a2 = Layer2(a1) # a2 is the final probability (e.g., 0.8)
12
13 # Optional: Make a binary prediction
```

```
14 | if a2 >= 0.5:
15 |     y_hat = 1
16 | else:
17 |     y_hat = 0
```

Example 2: Handwritten Digits (3-layer network)

This example shows how to stack multiple layers.

- **Input (Layer 0):** $\vec{x}$ (a vector of pixel intensity values)

```
1 | # Layer 1 (Hidden)
2 | Layer1 = Dense(units=25, activation='sigmoid')
3 | a1 = Layer1(x)
4 |
5 | # Layer 2 (Hidden)
6 | Layer2 = Dense(units=15, activation='sigmoid')
7 | a2 = Layer2(a1)
8 |
9 | # Layer 3 (Output)
10 | Layer3 = Dense(units=1, activation='sigmoid')
11 | a3 = Layer3(a2) # a3 is the final prediction
```

Note: The lab exercises will provide the code for loading the TensorFlow library and the pre-trained parameters (w and b) for each layer. The code above defines the *architecture* and executes the *forward propagation* steps.